

$$\underline{1-1. T(n) = 3T(n/2) + n^2, a=3, b=2}$$

$$C_{crit} = \log_2(3)$$

↑
critical growth rate
↓
1.585

$$f(n) = n^2$$

↑
extra work

↑
There are 3 recursive calls

↑
factor by which the problem shrinks (each list split in half)

$$2 > 1.585$$

$f(n)$ is polynomially larger than C_{crit} , so case 3 $T(n) = \Theta(n^2)$

Case 3 condition:

$$\underline{3 f(n/2) = 3 (n/2)^2 = \frac{3}{4} n^2, \frac{3}{4} n^2 \leq c n^2}$$

If $f(n)$ grows faster than C_{crit} term, think Case 3 — for $c = 3/4 < 1$

Case 2

If it grows at same rate:

Case 2

Slower: Case 1

$$\underline{1-2. T(n) = 7T(n/2) + n^2, a=7, b=2}$$

$$C_{crit} = \log_2(7) = 2.807 \quad f(n) = n^2 \quad \text{Case 1 condition:}$$

$$n^{C_{crit}} = n^{2.807} \quad 2.807 > 2 \quad \frac{7}{4} n^2 \geq c n^2$$

$n^{C_{crit}}$ is larger, so case 1

$c < 1$
 $T(n) = \Theta(n^{\log 7})$

1-3. $T(n) = 4T(n/2) + n^2$ $a=4, b=2$

$C_{crit} = \log_2(4)$ $f(n) = n^2$

$C_{crit} = 2$ $2 = 2$

$T(n) = \Theta(n^{\overset{C_{crit}}{2}}(\log n)) = \Theta(n^2(\log n))$

Case 2

1-4. $T(n) = 3T(n/4) + n \log n$

$C_{crit} = \log_4(3)$ $f(n) = n \log n$

$\log_4(3) = 0.792$ $n^{0.792} < n \log n$

Case 3: $T(n) = \Theta(n \log n)$

1-5. $T(n) = 4T(n/2) + \log n$

$C_{crit} = \log_2(4) = 2$ $f(n) = \log n$

$n^{C_{crit}} = n^2 > \log n$

↑
grows at
greater rate

$T(n) = \Theta(n^2)$

Case 1

1-6. $T(n) = T(n-1) + n$

The problem does not involve splitting the work by a constant factor, so Master Theorem does not apply.

1-7. $T(n) = 4T(n/2) + n^2 \log n$

$C^{\text{crit}} = \log_2(4) = 2$ rate of increase:
 $n^2 < n^2 \log n$
 $\uparrow$

$T(n) = \Theta(n^2 \log n (\log n))$ not case 3 because
 $T(n) = \Theta(n^2 (\log n)^2)$ it's larger by a logarithmic factor, not polynomially larger

1-8. $T(n) = 5T(n/2) + n^2 \log n$ Polynomically larger

$C^{\text{crit}} = \log_2(5) = 2.322$ $n^{2.322} > n^2 \log n$

Case 1: $T(n) = \Theta(n^{\log_2(5)})$

1-9. $T(n) = 3T(n/3) + n/\log n$

$C^{\text{crit}} = \log_3(3) = 1$ $n \frac{n}{\log n}$

Master Theorem doesn't apply since $n/\log n$ is not Polynomically smaller than n .

1-10. $T(n) = 2T(n/4) + c$

$C^{\text{crit}} = \log_4(2) = \frac{1}{2}$ $n^{\frac{1}{2}} > c n^0$
 $\downarrow$

Case 1: $T(n) = \Theta(n^{\log_4(2)})$

1-11. $T(n) = T(n/4) + \log n$ Same Polynomial
 $C^{\text{crit}} = \log_4(1) = 0$ $\underset{1}{n^0} = n^0 \log n$

case 2: $T(n) = \Theta(f(n) \log n) = \Theta(\log n (\log n)) = \Theta((\log n)^2)$

1-12. $T(n) = T(n/2) + T(n/4) + n^2$

In master theorem form: $T(n) = aT(n/b) + f(n)$
 the only type of recursive subproblem is $aT(n/b)$. So all a subproblems have to be size n/b .

In the given problem there are 2 different subproblem sizes: $n/2$ & $n/4$. There is no single a & b so master theorem doesn't apply.

1-13. $T(n) = 2T(n/4) + \log n$

$C^{\text{crit}} = \log_4(2) = \frac{1}{2}$ $n^{\frac{1}{2}} > n^0 \log n$

Case 1: $T(n) = \Theta(n^{\frac{1}{2}})$

1-14. $T(n) = 3T(n/3) + n \log n$

$C^{\text{crit}} = \log_3(3) = 1$ $n^1 = n^1 \log n$

Case 2: $T(n) = \Theta(n \log n (\log n)) = \Theta(n (\log n)^2)$

1-15. $T(n) = 8T((n - \sqrt{n})/4) + n^2$

$T\left(\frac{(n - \sqrt{n})}{4}\right) \rightarrow T\left(\frac{n}{4} - \frac{\sqrt{n}}{4}\right)$

Cannot directly be matched to the form $aT(n/b)$.
Master theorem does not apply

1-16. $T(n) = 2T(n/4) + \sqrt{n}$ (also $n^{\frac{1}{2}}$)

$c^{\text{crit}} = \log_4(2) = \frac{1}{2}$ $n^{\frac{1}{2}} = n^{\frac{1}{2}}$

Case 2: $T(n) = \Theta(n^{\frac{1}{2}} \log n)$

1-17. $T(n) = 2T(n/4) + n^{0.51}$

$c^{\text{crit}} = \log_4(2) = \frac{1}{2}$ $n^{\frac{1}{2}} < n^{0.51}$

Case 3: $T(n) = \Theta(n^{0.51})$

1-18. $T(n) = 16T(n/4) + n!$ ← grows much faster than any exponent

Case 3: $T(n) = \Theta(n!)$

1-19. $T(n) = 3T(n/2) + n$

$c^{\text{crit}} = \log_2(3) = 1.585$ $n^{1.585} > n^1$

Case 1: $T(n) = \Theta(n^{\log_2(3)})$

1-20. $T(n) = 4T(n/2) + cn$

$c^{\text{crit}} = \log_2(4) = 2 \quad n^2 > n'$

Case 1: $T(n) = \Theta(n^2)$

1-21. $T(n) = 3T(n/3) + \frac{n}{2}$

$c^{\text{crit}} = \log_3(3) = 1 \quad n' = n'(\frac{1}{2})$

Case 2: $T(n) = \Theta(n \log n)$

1-22. $T(n) = 4T(n/2) + n/\log n$

$c^{\text{crit}}: \log_2(4) = 2 \quad n^2 > n'(\frac{1}{\log n})$

Case 1: $T(n) = \Theta(n^2)$

1-23. $T(n) = 7T(n/3) + n^2$

$c^{\text{crit}}: \log_3(7) = 1.771 \quad n^{1.771} < n^2$

Case 3: $T(n) = \Theta(n^2)$

1-24. $T(n) = 8T(n/3) + 2^n$

$c^{\text{crit}}: \log_3(8) = 1.893 \quad n^{1.893} < 2^n$

Case 3: $T(n) = \Theta(2^n)$

1-25. $T(n) = 16T(n/4) + n$

Crit: $\log_4(16) = 2 \quad n^2 > n'$

Case 1: $T(n) = \Theta(n^2)$

AI Usage: checking work, writing docstrings, & writing simple file import functions